

Review Article

Machine learning in ecology: Current status and future directions in the big data era



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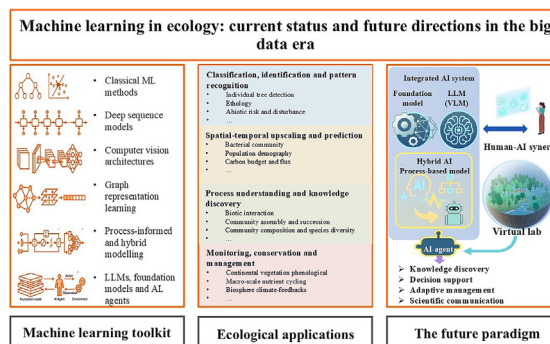
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HIGHLIGHTS

- We synthesize three decades of machine learning advances in ecology and global biology.
- Machine learning supports ecological pattern detection, prediction, monitoring, and inference.
- Emerging artificial intelligence has the potential to enable adaptive, human-guided ecological workflows.

GRAPHICAL ABSTRACT



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ABSTRACT

Ecology is entering a data-rich era driven by rapid advances in sensor technologies and the expansion of environmental observations across spatial and temporal scales. In parallel, machine learning (ML) has been increasingly adopted as a powerful tool for analysing complex ecological systems, where interactions among organisms and their environments are often nonlinear, heterogeneous and scale-dependent. Over the past three decades, ML applications in ecology and global biology have expanded rapidly, supporting advances in pattern detection, prediction, monitoring and process understanding, while still facing major challenges in data quality, sampling bias, interpretability, uncertainty quantification and causal inference. Emerging developments in embodied Artificial Intelligence (AI), foundation models, large language models and intelligent agent systems now create new opportunities for ecological research by enabling adaptive observation, improved integration of heterogeneous data and more efficient analytical workflows. These advances are beginning to steer the field towards an AI-assisted research paradigm, in which ecologists supervise coordinated AI systems for data collection, integration, analysis and modelling, while providing interpretation, validation and iterative feedback. Realizing the full potential of such human-AI synergies in ecological studies will require closer integration between ecological knowledge and advanced AI methods, as well as sustained collaboration among ecologists, data scientists and technology developers.

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1. Introduction

Ecology examines interactions among organisms and their abiotic environment, and how this knowledge can address environmental challenges (Odum, 1971; Sutherland et al., 2013; Zettlemoyer et al., 2023). Over the past four decades, its scope has expanded from classical plant- and population-focused studies to a broader framework that includes microscale processes, macroscale patterns, and anthropogenic influences, driven by concerns over climate change, biodiversity loss, ecosystem degradation, and sustainability (McCallen et al., 2019). Global change is exposing and amplifying complex ecological interactions across multiple spatial and temporal scales, making ecosystem dynamics increasingly challenging to observe, understand, and predict (Clark et al., 2001; Oliver and Roy, 2015; Huang et al., 2019). Meanwhile, advances in modern techniques have enabled ecologists to generate large volumes of heterogeneous data, including genetic, trait, distributional, demographic, movement, acoustic, and environmental information (Luo et al., 2011). As data increase in scale and complexity, ecology has shifted toward a multidisciplinary and data-intensive science (Hampton et al., 2013; McCallen et al., 2019). This influx of massive, heterogeneous datasets offers unprecedented opportunities but requires analytical tools capable of untangling intricate ecological patterns. However, traditional statistical and process-based approaches are often challenged by high dimensionality, nonlinearity, and complex interactions (Olden et al., 2008). Advances in computing, algorithms, and data availability have enabled machine learning (ML) to emerge as a powerful tool for uncovering patterns, identifying drivers, and improving predictions across scales (Christin et al., 2019).

Over recent decades, ML has become integral to the ecological data-model continuum, supporting prediction, feature extraction, dimensionality reduction, causal inference, data assimilation, uncertainty quantification, and integration of heterogeneous data. These applications span diverse methods, from neural networks to ensemble and hybrid approaches, and multiple ecological scales, including molecular and genetic studies (Kalyakulina et al., 2020), individual-level applications such as plant phenotyping (Jiang and Li, 2020) and animal behavior monitoring (Jeantet et al., 2021), and population- and community-level research such as species distribution mapping and outbreak prediction (Deneu et al., 2021; Ramazi et al., 2021). At the ecosystem and global scale, ML is increasingly applied to evaluating and predicting carbon and water dynamics, identifying ecosystem-climate interactions, and detecting the early signals of tipping points (Bury et al., 2021). The expanding role of ML across ecological research activities, from ecological observation and monitoring to modeling, forecasting, and decision support, highlights the need for a unified synthesis of methods, applications, and future opportunities in the big-data era.

Despite rapid progress and broad applications in areas such as biodiversity monitoring, conservation, water management, soil mapping, and ecosystem services (Shen, 2018; Wadoux et al., 2020; Stowell, 2022; Manley et al., 2022; Tuia et al., 2022), key challenges remain. Ecology requires not only accurate prediction but also mechanistic understanding to enhance generality and interpretability. While ML improves model fit, it increases the risk of overfitting, making validation and bias-variance tradeoffs critical (Pichler and Hartig, 2023). Traditional experiments provide mechanistic insights but are often limited in scale, whereas process-based models encode known mechanisms yet struggle to fully utilize high-dimensional data. Bridging data-driven and mechanism-based approaches is a central challenge (Jin et al., 2026). Meanwhile, foundation models are emerging as a promising direction for ecological research, with the potential to support knowledge transfer, multimodal integration, and large-scale ecological analysis as their applications continue to develop (Zhu et al., 2026). Here, we synthesize ML applications across ecological domains and examine how advances such as foundation models and knowledge-guided ML are reshaping the field toward a unified mechanism and data-driven paradigm. We then outline: (1) a brief introduction to ML methodology in ecology; (2) a

thematic review of ML applications in ecology, covering classification and identification, spatial-temporal prediction, process understanding, and conservation and management; (3) a discussion on challenges such as integrating ML with process-based models for insightful ecological studies; and (4) outlooks and approaches to move forward.

2. Characteristics and evolution of ML methods in ecology

Machine learning has expanded rapidly in ecological applications over the past three decades (Fig. 1). This development benefits from efforts to overcome the limitations of traditional parametric statistics which faces challenges in addressing high dimensional, nonlinear, and complex ecological processes. Classical machine learning methods used relatively simple models with limited numbers of parameters to learn nonlinear relationships from a fixed set of ecological predictors. They were therefore well suited to structured environmental data and could often achieve stable performance with moderate sample sizes. As ecological datasets became larger and more diverse, deep learning extended the capacity of machine learning through neural networks with more layers and trainable parameters. These deeper architectures can learn complex patterns from large datasets and high-dimensional inputs such as images, long-term time series, acoustic recordings and spatial grids. Compared with classical ML, deep learning usually requires more data and computation, but its larger model capacity allows performance to continue improving as data scale and complexity increase. More recent developments, including graph learning, process-informed modeling, foundation models and Artificial Intelligence (AI) agents, have further extended machine learning to ecological interactions, multimodal observations, ecological knowledge and complex scientific workflows.

Based on the major methodological developments and their ecological relevance, we categorize ML approaches commonly used in ecological research into six broad groups: (1) classical ML methods, (2) deep sequence models, (3) computer vision architectures, (4) graph representation learning, (5) process-informed and hybrid modeling, (6) large language models, foundation models and AI agents. Table S1 in the Supplementary materials summarize the main characteristics, inputs and representative ecological applications of these methods, whereas Section 3 discusses their applications in greater detail.

2.1. Classical ML methods

Classical ML methods remain widely used in ecology because they provide robust performance on structured environmental datasets while requiring relatively limited training data and offering moderate interpretability. Representative methods include maximum entropy (MaxEnt), decision trees and support vector machines (Cortes and Vapnik, 1995; Berger et al., 1996; Loh, 2011). Unlike traditional parametric models, these methods learn relationships directly from data without specifying the underlying functional form in advance. Differences among algorithms mainly lie in how they use features to build prediction functions. MaxEnt models estimate probability distributions through weighted combinations of feature functions under expectation constraints (Berger et al., 1996). Decision trees recursively partition data into local regions based on feature thresholds (Loh, 2011), while kernel-based methods such as support vector machines (SVM) construct decision functions in high-dimensional feature spaces through kernel mappings.

Building on feature-based approaches, ensemble learning improves performance by combining multiple base learners. Representative approaches include random forests, which reduce prediction variance through bagging, and boosting algorithms such as gradient boosting machines (GBM), XGBoost and LightGBM, which iteratively improve predictive accuracy (Breiman, 1996; Friedman, 2001; Chen and Guestrin, 2016; Ke et al., 2017). Owing to their robustness, moderate data requirements and relatively good interpretability, classical ML

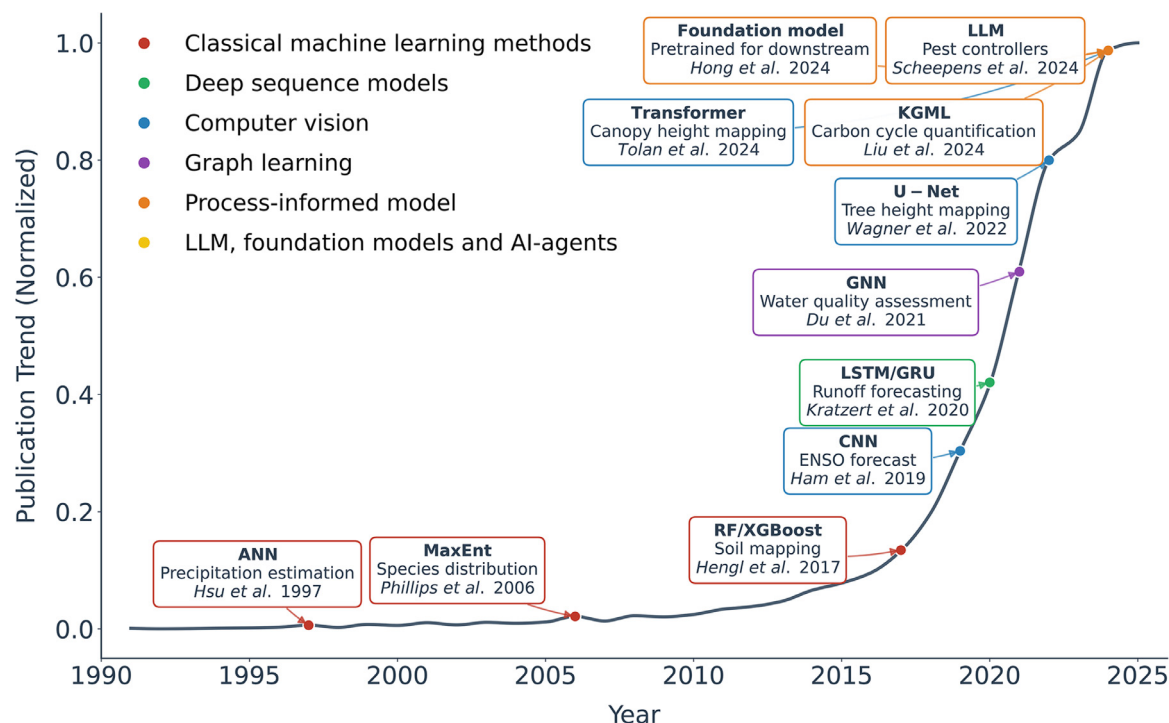


Fig. 1. Publication trends and representative applications of machine learning (ML) in ecological research. Bibliometric data were obtained from the Web of Science Core Collection for the period 1990–2025. Details of search queries, inclusion criteria and data processing are provided in the Supplementary materials. Annotated examples correspond to highly cited studies selected to represent typical ecological applications of each major ML approach. ANN, artificial neural networks; MaxEnt, maximum entropy; RF/XGBoost, Random Forest/extreme gradient boosting; CNN, convolutional neural networks; LSTM/GRU, long short-term memory networks/gated recurrent units; GNN, graph neural networks; KGML, knowledge-guided machine learning; LLM, large language models.

methods remain widely used in ecology, particularly for structured environmental datasets (Yu et al., 2025).

2.2. Deep sequence and temporal dynamics

Ecological dynamics are strongly shaped by historical environmental conditions, whose effects can persist long after they occur (Bruce et al., 2007). Deep sequence models are designed to learn temporal dependencies from ordered observations. Through recurrent structures, attention mechanisms or other sequence-based architectures, they use information from earlier time steps to model current and future states (Benidis et al., 2022). By combining multilayer representations with recurrent, convolutional or attention-based mechanisms, deep sequence models can capture nonlinear patterns across different temporal scales, including short-term variation, seasonal cycles and long-range dependencies (Lim and Zohren, 2021). This makes them well suited to ecological forecasting and the analysis of ecosystem dynamics.

Representative deep sequence models include recurrent neural networks (RNNs), long short-term memory networks (LSTMs), gated recurrent units (GRUs), and their variants (Hochreiter and Schmidhuber, 1997; Cho et al., 2014). Compared with standard RNNs, LSTM and GRU use gating mechanisms to retain relevant information over long sequences. EA-LSTM further introduces an attention mechanism that gives different weights to variables and temporal sub-windows, helping the model identify the historical information most useful for prediction (Li et al., 2019). These models have been used to predict soil moisture, carbon fluxes and gross primary productivity, and, when combined with explainable methods, to examine ecological memory and biome-specific controls on ecosystem dynamics (Ma et al., 2026).

More recently, Transformer architectures have been used for ecological time series by modelling long-range dependencies through self-attention. State space models (SSMs), represented by Mamba, provide

another promising direction by improving computational efficiency through selective state updates (Gu and Dao, 2023). Although SSMs have not yet been widely applied to ecological or biogeochemical time series, they have shown promise in related fields such as functional genomics (Fazzari et al., 2025), highlighting their potential for future ecological applications.

2.3. Computer vision and spatial representation

Computer vision architectures are designed to learn hierarchical representations from spatially structured ecological data, particularly remote sensing imagery and gridded observations. In spatial modeling, many ecological variables exhibit strong structural properties, including local similarity, multiscale heterogeneity, and spatial autocorrelation. Convolutional neural networks (CNNs) are well suited to capture these patterns through local receptive fields and shared kernels, enabling efficient extraction of multiscale spatial features from raw inputs (LeCun et al., 1998). Consequently, CNNs have become the widely applied architecture for analysing ecological imagery, digital elevation models, and other spatial datasets, substantially improving predictive performance in applications such as digital soil mapping (Padarian et al., 2019).

Subsequent model development has focused on improving deep network trainability and capturing long-range spatial dependencies. Residual networks (ResNet) introduce identity mappings that alleviate vanishing gradients and support deeper architectures (He et al., 2016). Building upon CNNs, encoder–decoder architectures represented by U-Net further improve dense prediction by combining hierarchical feature extraction with skip connections that preserve fine spatial details during decoding (Ronneberger et al., 2015). This design has become widely used for ecological image segmentation, including individual tree crown delineation, habitat mapping, and land-cover classification (Freudenberger et al., 2022).

Although CNN-based architectures effectively capture local spatial patterns, modelling long-range spatial dependencies remains challenging. Vision Transformers (ViTs) address this limitation through self-attention mechanisms that directly model interactions between distant spatial locations, improving representations of large-scale ecological structures (Dosovitskiy et al., 2020). More recently, self-supervised and contrastive learning approaches have further reduced dependence on labelled data by pretraining models on large collections of unlabeled imagery, enabling more generalizable spatial representations for downstream ecological applications (Chen et al., 2020; He et al., 2020).

2.4. Graph learning and topological connectivity

Many ecological processes are organized by connectivity rather than geographic proximity. Species interactions, hydrological flows, landscape connectivity, and upstream–downstream dependencies are often determined by ecological networks rather than regular spatial grids (Erős and Lowe, 2019). Graph embedding provides a way to represent ecological networks as connected entities, with nodes denoting species and edges describing their potential interactions (Strydom et al., 2023). Building on this representation, graph neural networks learn from both node attributes and network structure, allowing ecological interactions and other relational patterns to be incorporated into prediction (Scarselli et al., 2009; Battaglia et al., 2018).

Unlike CNN, which aggregates information from fixed local neighborhoods, GNNs propagate information through graph connectivity using message passing (Gilmer et al., 2017). This enables models to capture long-range dependencies, irregular interactions, and cross-scale processes while naturally integrating multisource information into node representations (Hamilton et al., 2017; Wang et al., 2019; Abdi et al., 2025). Consequently, GNNs have become increasingly useful for ecological applications involving species interaction networks, river systems, habitat connectivity, and landscape processes (Wu et al., 2025; Stone et al., 2026).

Recent developments have further extended GNNs to dynamic systems. Spatiotemporal GNNs combine graph propagation with temporal modelling, allowing node states to evolve according to both their own history and interactions with neighbouring nodes (Li et al., 2017; Guo et al., 2019). A representative example is GraphCast, which represents atmospheric dynamics using hierarchical graph structures instead of regular grids, enabling efficient long-range information propagation for weather forecasting (Lam et al., 2023). Such developments highlight the potential of graph-based learning for modelling complex ecological and Earth system dynamics.

2.5. Process-informed and hybrid modeling

Although modern ML models have achieved high predictive performance, they primarily learn statistical relationships from observations and do not explicitly incorporate ecological mechanisms. This limitation becomes particularly important when observations are sparse, noisy, or collected under changing environmental conditions, where learned correlations may not remain valid outside the training domain (Reichstein et al., 2019; Beucler et al., 2020). Consequently, improving model consistency with ecological processes while maintaining predictive flexibility has become an important direction in ecological ML.

Process-informed and hybrid modeling address this challenge by integrating domain knowledge into ML (Reichstein et al., 2019). Rather than relying solely on data-driven correlations, these approaches improve model reliability by integrating established ecological understanding into different stages of model development (Jin et al., 2026). Several strategies have emerged to achieve this integration. Ecological knowledge can be embedded into model structures by explicitly representing key system components and their interactions, allowing ML to preserve known ecological mechanisms rather than learning relationships solely from data (Karpatne et al., 2017). It can also be incorpo-

rated as learning constraints by introducing governing equations, conservation laws, stoichiometric relationships, and empirical ecological principles into model architectures or objective functions, thereby improving physical consistency and generalization under sparse observations (Raissi et al., 2019; Beucler et al., 2020). Hybrid modelling further combines ML with process-based models to exploit their complementary strengths, while differentiable modelling enables joint optimization of process parameters and neural network components through end-to-end learning (Shen et al., 2023). These strategies have been successfully applied to ecological prediction, for example, improving carbon cycle modelling through knowledge-guided machine learning (Liu et al., 2024; Fang et al., 2026).

Despite their different implementations, these approaches share the common objective of combining mechanistic understanding with ML. By improving interpretability, physical consistency, and extrapolation ability, process-informed and hybrid models provide an important bridge between traditional process-based ecology and modern artificial intelligence.

2.6. Large language model, foundation models and AI agents

Large language models (LLMs) are generative models based on transformer architectures and pre-trained on massive text corpora. They process text as sequences of tokens, such as words, subwords, or characters, and are increasingly used in ecology through prompt-based adaptation to specific tasks, such as employing few-shot prompting to extract plant functional traits from unstructured biological texts with mean precision of 90.8 % (Domazetoski et al., 2025). Given a prompt, the model encodes the input and predicts subsequent tokens to generate coherent text (Hsiao and Hung, 2022). Rather than memorizing answers, LLMs learn statistical regularities in language, allowing them to generalize across tasks through different prompts (Brown et al., 2020). Transformer self-attention enables the model to capture both local context and long-range dependencies within a sequence (Vaswani et al., 2017). Building on this capability, vision-language models (VLMs) align images and text in a shared representation space, enabling joint analysis of multimodal ecological information such as literature, remote sensing imagery, maps, and scientific figures (Radford et al., 2021).

Foundation models further extend this paradigm by pretraining on diverse large-scale datasets to learn general-purpose representations rather than solving a single task (Bommasani et al., 2021). Through self-supervised learning, they capture spatial, temporal, and cross-modal patterns that can be transferred across multiple downstream ecological applications. Consequently, the pretrained model can support prediction, classification, segmentation, retrieval, and other tasks with minimal task-specific adaptation. Recent studies suggest that foundation models can jointly represent vegetation dynamics, climate drivers, and land-surface processes, providing a promising framework for multitask ecological analysis (Lu et al., 2024; Zhu et al., 2026).

AI agents build on pretrained models by adding goal-directed decision making, planning, and execution (Wang et al., 2024). Instead of generating text alone, agents can combine multimodal observations with external tools, databases, models, and code environments to perform complex tasks (Wei et al., 2022). In ecology, such systems could connect directly to meteorological stations, flux towers, unmanned aerial vehicles, or satellite data streams to support environmental monitoring, data processing, anomaly detection, process interpretation, and scenario analysis (Besson et al., 2022).

3. Applications of ML in ecology

The integration of ML into ecological research has sparked a paradigm shift, unlocking new possibilities for understanding and managing ecosystems in ways that were previously unimaginable. From species identification to ecosystem dynamics, ML is reshaping how

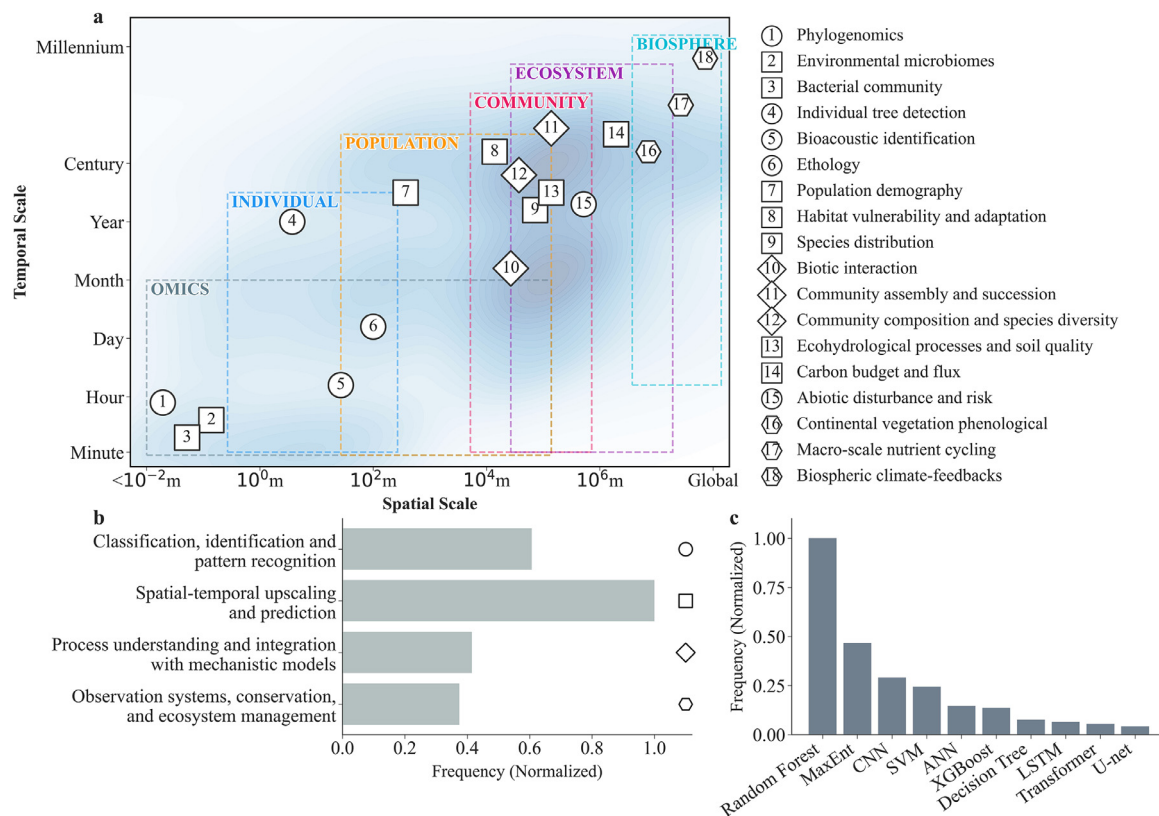


Fig. 2. Spatiotemporal distribution and functional landscape of machine learning (ML) in ecological research. **(a)** Spatiotemporal domains of ML applications. Eighteen representative ecological tasks are shown, defined as the three most frequently reported applications within each of six ecological hierarchy levels (Omics, Individual, Population, Community, Ecosystem and Biosphere). Background contours indicate the kernel density estimation of study distribution. Marker shapes denote primary research purpose: classification, identification and pattern recognition (circles); spatial-temporal upscaling and prediction (squares); process understanding and knowledge discovery (diamonds); and monitoring, conservation and management (hexagons). Details of literature compilation, classification criteria and scale assignment are provided in Supplementary materials. **(b)** Relative frequency of ML applications by research purpose. Bars represent proportions normalized to the most frequent research purpose. **(c)** Relative usage of ML algorithms across the literatures. Frequencies are normalized to the most commonly used method. CNN, convolutional neural networks; SVM, support vector machine; ANN, artificial neural networks; LSTM, long short-term memory.

we monitor biodiversity, predict environmental changes, and make informed decisions in conservation. In this section, we review ML applications in ecology from a thematic perspective, focusing on how ML supports different stages of ecological inquiry and decision-making across multiple spatial and temporal scales (Fig. 2a). Specifically, we organize existing studies into four broad but interconnected application domains: (1) classification, identification, and pattern recognition; (2) spatial-temporal upscaling and prediction; (3) process understanding and knowledge discovery; and (4) monitoring, conservation and management. As visualized in the spatiotemporal and functional landscape of ecological ML applications, these four domains span across six fundamental ecological hierarchies (from omics to the biosphere), addressing diverse tasks tailored to specific spatiotemporal constraints (Fig. 2a, b). Together, these domains illustrate how ML contributes not only to improved ecological monitoring and prediction, but also to deeper ecological insight and more informed environmental decision-making.

3.1. Classification, identification and pattern recognition

Many fundamental ecological questions depend on extracting objects, states, and recurring patterns from complex observational data. Classification is widely used in ecology, especially in remote sensing, where ML supports high-resolution land cover mapping, long-term forest change monitoring, and delineation of ecological or climatic zones across broad spatial scales (Metzger et al., 2013; Vancutsem et al., 2021; Brown et al., 2022). These products standardize land-surface proper-

ties, facilitating ecosystem comparison and integration into process-based models. ML also supports the derivation of plant functional types (PFTs) for land surface models from observational data, including underrepresented systems such as wetlands and aquatic environments (Dronova et al., 2012). However, ecological classification requires careful consideration of representation, because discrete classes may overlook continuous variation and mixed functional states common in real ecosystems (Cranko Page et al., 2024). Its value therefore depends not only on predictive accuracy, but also on whether the categories retain ecological meaning for the question being addressed.

Beyond broad categorization, identification focuses on detecting specific ecological entities, states, or targets from complex backgrounds (Fassnacht et al., 2016; Weinstein et al., 2020; Wang et al., 2021; Bastarrika et al., 2024). In many cases, this means separating targets from complex surrounding structures, such as distinguishing individual trees in dense forest canopies or organisms that are partially occluded in heterogeneous scenes (Lang et al., 2023). Deep learning has substantially improved this capacity. For example, individual tree crowns can be delineated from high-resolution LiDAR point clouds despite overlapping structures, enabling estimation of tree height, canopy structure, biomass, and rare extremes such as exceptionally tall trees (Ferraz et al., 2016). Similar feature-learning approaches are also widely used in large observational datasets. In camera trap networks, ML enables standardized processing of large image collections, supporting analyses of activity patterns, seasonal dynamics, and species co-occurrence (Wang, 2019; Willi et al., 2019). Applied to crowd-sourced platforms such as iNatural-

ist and BirdNET, related methods make it possible to use geo-referenced image and acoustic records to examine species turnover and distributional shifts across environmental gradients (Van Horn et al., 2018; Kahl et al., 2021).

Pattern recognition extends beyond discrete labels and target detection to extracting structured variation from ecological data, especially when signals are continuous in space or time. In spatial datasets, ML can identify gradients in nutrient limitation, microbial strategies, or disturbance regimes, allowing heterogeneous landscapes to be represented as continuous or clustered ecological structures rather than fixed classes (Du et al., 2020; Senf and Seidl, 2021; Piton et al., 2023). In temporal data, similar approaches are applied to continuous sensor records. For example, bio-logging data can be segmented into behavioral states such as foraging or migration, enabling reconstruction of activity budgets and movement strategies over time (Bergen et al., 2022; Nathan et al., 2022). At longer time scales, ML analysis of remote sensing data supports detection of phenological transitions, including green-up and senescence, which can be linked to climate variability and ecosystem productivity (Huang et al., 2023). These approaches are especially valuable in systems with continuous data streams or limited accessibility, such as polar regions (Borowicz et al., 2018).

3.2. Spatial-temporal upscaling and prediction

By integrating heterogeneous environmental data and capturing complex nonlinear relationships, ML provides an important tool for reconstructing continuous ecological states from sparse observations (Hengl et al., 2017). Because ecological observations are uneven in space and time, reliable extrapolation and cross-scale inference remain major challenges. Traditional statistical approaches, such as regression kriging, represent ecological variables as a linear combination of environmental covariates plus a spatially autocorrelated residual (Goovaerts, 1999). Such frameworks rely on predefined functional relationships between environmental drivers and ecological responses, limiting their flexibility in representing complex interactions among multiple factors.

ML can learn high-dimensional nonlinear mappings and has been widely applied to reconstruct large-scale ecological variables. For example, studies have quantified global soil inorganic carbon storage across soil profiles and highlighted its sensitivity to environmental change, especially nitrogen-driven acidification (Huang et al., 2024). Similar approaches have been extended to vegetation and community properties. By integrating optical remote sensing with gap-filled observations from the global plant trait database (TRY), random forest has been used to generate global leaf trait maps at 500 m resolution, including specific leaf area, leaf dry matter content, and leaf nutrient concentrations (Moreno-Martínez et al., 2018). These studies show that ML, supported by multisource data, can effectively extrapolate from point observations to continuous spatial distributions. Random forest remains one of the most widely used approaches for spatial interpolation and upscaling (Fig. 2), but its training commonly assumes independent and identically distributed samples and therefore neglects spatial autocorrelation (Ploton et al., 2020). When training data are spatially clustered, models may learn region-specific relationships rather than general ecological mechanisms. This problem becomes more severe in poorly sampled regions such as Siberia and tropical rainforests, where predictions may become unstable or ecologically implausible (Meyer and Pebesma, 2022). Recent studies have therefore begun to incorporate explicit spatial modeling approaches, such as convolutional and graph neural networks, or to combine ML with process-based models to improve generalization.

Beyond static variables, ML has also been widely applied to the spatiotemporal upscaling of ecosystem fluxes, which is essential for understanding land-atmosphere interactions and carbon sink-source dynamics (Jung et al., 2017). Unlike static properties, ecological fluxes vary over time and require joint treatment of spatial and temporal scales. Models trained on eddy covariance observations can upscale site-level

carbon, water, and energy fluxes to regional and continental scales (Tramontana et al., 2016). Early studies showed that artificial neural networks could generate continuous flux estimates, constrain the magnitude of the European carbon sink and identify phenological variation as a key driver (Papale and Valentini, 2003). More recent work has integrated geostationary satellite observations to estimate carbon fluxes at half-hourly resolution, showing that tree-based ensemble methods can reasonably predict gross primary productivity and ecosystem respiration and thus provide near-real-time constraints on sub-daily carbon cycling (Yang et al., 2022). At the global scale, multisource ML models combining satellite and airborne observations have generated spatially explicit products at about 30 m resolution for vegetation cover, photosynthesis, and canopy structure (Moreno-Martínez et al., 2018; Ma et al., 2021; Lang et al., 2023). However, most current multisource fusion studies still focus on single variables, whereas multitask learning remains underemployed. The main advantage of multitask learning is the ability to capture the inherent relationships among ecological variables, improving physical consistency and reducing problems such as violations of mass conservation or stoichiometric imbalance. Recent advances in foundation models may provide a unified framework for multitask learning, but their ecological applications are still in their infancy and require further investigation.

ML can also be used to identify nonlinear thresholds in ecosystem dynamics and provide early warning of ecological change. Compared with methods based on single statistical indicators such as variance change, ML can learn more complex dynamic patterns from time series and detect signals such as critical slowing down as systems approach tipping points (Bury et al., 2021). Deep neural networks trained on theoretical models, such as EWSNet, can identify critical transitions in empirical time series and distinguish whether changes are catastrophic or gradual (Deb et al., 2022; Lenton et al., 2024). Related approaches have also been applied to predict extreme climate events such as heatwaves and droughts (Ham et al., 2019; Chattopadhyay et al., 2020).

3.3. Process understanding and knowledge discovery

Although ML has achieved remarkable success in ecological prediction, its applications have traditionally focused on pattern recognition rather than process understanding. As a result, many models struggle with causal inference because they rely primarily on statistical associations rather than explicit mechanisms (Arif and MacNeil, 2022). However, ML is increasingly evolving toward extracting underlying ecological rules. Methods such as SHAP and Integrated Gradients estimate feature contributions to predictions, while local surrogate models such as LIME approximate model behavior in restricted regions (Ribeiro et al., 2016; Lundberg and Lee, 2017; Sundararajan et al., 2017). Global interpretation tools, including partial dependence and accumulated local effects plots, further characterize average feature influences (Friedman, 2001; Apley and Zhu, 2020). Together, these approaches allow ecologists to decompose model behavior, quantify driver importance, and identify nonlinear response thresholds. Beyond improving model interpretability, ML is increasingly contributing to process understanding and knowledge discovery. One important direction is causal inference, where causal ML frameworks can disentangle confounding effects and infer directed relationships from observational data (Runge et al., 2023). Process understanding can also emerge when domain knowledge is combined with information extracted from unstructured data. For example, ML has been used to infer species interactions in ecological networks by learning trait-matching relationships between interacting species, allowing ecologists to predict plant-pollinator interactions and identify ecologically meaningful interaction rules from observational data (Pichler and Hartig, 2023).

In addition to data-driven inference, process understanding can be enhanced by integrating ML with process-based models. These models represent ecological mechanisms through coupled equations but rely on parameters that are often uncertain and difficult to constrain

(Luo and Schuur, 2020). ML can learn spatially explicit parameterizations from data, improving representation of ecological heterogeneity. For instance, replacing static parameters with ML-derived trait maps allows models to capture trait plasticity and environmental responses (Moreno-Martínez et al., 2018). Similarly, deep learning has been applied to interpolate spatially heterogeneous parameter fields for the Community Land Model version 5 (CLM5) by assimilating 25,444 vertical soil profiles, significantly reducing geographical biases (Tao et al., 2020). ML also supports process understanding in complex simulation frameworks. For example, combining eco-evolutionary simulations with supervised learning can reveal mechanisms underlying biodiversity patterns (Overcast et al., 2021). In addition, ML can emulate computationally expensive models or replace poorly understood components, forming hybrid or gray-box systems that retain physical constraints while improving flexibility (Reichstein et al., 2019; Li et al., 2023). These approaches represent a shift from pure prediction toward mechanistic understanding.

3.4. Monitoring, conservation and management

Ecological monitoring aims to track ecosystem dynamics and environmental changes through space and time. Approaches such as ML-based segmentation methods provide a convenient approach by enabling extraction of ecological features from repeated observations and integrating them with other types of data to support large-scale monitoring (Reichstein et al., 2019). Conventional field surveys are limited by sparse spatial sampling and low temporal frequency, whereas medium-to low-resolution satellite data, including 30 m imagery, are often insufficient to detect fine-scale changes such as scattered tree mortality or localized canopy degradation (Senf et al., 2017). Semantic and instance segmentation models can automatically identify standing deadwood from centimeter-scale UAV and aerial orthophotos with high precision using open high-resolution imagery, as demonstrated by tools such as deadtrees.earth (Mosig et al., 2026). These observations can then serve as training references and be integrated with multispectral satellite products that provide broad spatial coverage and temporal continuity, such as Sentinel and Landsat, enabling joint mapping and monitoring across scales (Ryu et al., 2019). By combining fine-resolution observations with broad-coverage remote sensing, this framework overcomes the limitations of high laboring requirements of monitoring and provides a basis for continuous and quantitative assessment of ecosystem anomalies at large scales (Turner, 2014).

This high-quality, multiscale observation capacity supports broad applications of ML in wildlife conservation and ecosystem assessment. ML has been applied to conservation status assessment, forest disturbance monitoring, automated species surveys, coral reef monitoring, and invasive species tracking (Pelletier et al., 2018; Popa-Báez et al., 2021; Rumelt et al., 2021; Sahani and Ghosh, 2021). In addition, reinforcement learning frameworks can improve long-term conservation planning by dynamically balancing biodiversity benefits and costs, outperforming static approaches while producing interpretable priority maps (Lapeyrolerie et al., 2022; Silvestro et al., 2022).

Building on conservation assessment, ML can also support a transition from passive monitoring to active ecosystem management. In restoration, ML models can jointly analyze environmental conditions, species requirements, and human pressures to inform site selection and intervention strategies, such as targeted reforestation or invasive species control (Keith et al., 2023). The integration of ML with real-time sensing and the Internet of Things further enables fine-scale adaptive management. End-to-end systems that combine low-cost sensors, drones, and ML algorithms can support continuous monitoring in both natural and agricultural landscapes. A prominent example is FarmBeats, which uses sensor networks and ML-based analytics to improve resource efficiency in agricultural systems (Vasishth et al., 2017). In wildlife management, combining GPS tracking data with ML can improve predictions of animal movement and habitat accessibility, thereby informing tar-

geted interventions such as resource deployment for feral animal control (Wang, 2019; Casazza et al., 2023). With continued advances in the Internet of Things, foundation models, and AI agents, these capabilities may further develop into adaptive management systems that integrate sensing, assessment, and intervention with reduced human input.

4. Key challenges

Although ML has been increasingly used in ecological research, several methodological and practical barriers still limit its reliability and scientific interpretability. These challenges are not independent: limitations in data quality and coverage propagate into issues of generalization and robustness, which in turn affect the robustness of models and the extent to which ML outputs can support mechanistic inference or decision-making. This section summarizes key challenges in ecological ML applications, focusing on data constraints, scale transfer and out-of-distribution behavior, interpretability and trust, the difficulty of causal inference and uncertainty quantification.

4.1. Data quality, imbalance and heterogeneity

The performance of ML in ecological research is constrained by data quality and availability. Although remote sensing provides broad spatial coverage, datasets are often affected by environmental noise and sensor limitations. Supervised models are particularly sensitive to such issues because high-quality labels require intensive expert validation. For example, when mapping soil carbon fractions such as mineral-associated organic carbon, models are limited by sparse field measurements or global soil spectral libraries that contain fewer than 10,000 reliable geo-referenced profiles (Ren et al., 2024). These limited observations were typically extrapolated across large spatial grids, allowing missing data and label inconsistencies to propagate into predictive uncertainty.

Data imbalance further restricts model generalization because ecological observations are often unevenly distributed across taxa, regions, and environmental conditions. In wildlife monitoring, model performance strongly depends on sample size. Schneider et al. (2020) showed that recall rates for species with fewer than 500 training images remain low and unstable, whereas performance improves substantially when sample sizes exceed 1,000 images. This problem extends beyond class imbalance to the representativeness of ecological observations. Many monitoring networks are disproportionately distributed across accessible regions or well-studied ecosystems, leaving large geographical and environmental domains underrepresented (Alton, 2020; Pallandt et al., 2022). Consequently, models may learn biases associated with sampling effort rather than general ecological relationships, limiting their transferability to poorly sampled regions (López-Blanco et al., 2024). Together, these limitations reduce the representativeness of ecological training data, restricting robust generalization across diverse ecosystems and potentially biasing large-scale ecological predictions.

The heterogeneity of ecological data also complicates model development. Predictive ecology increasingly relies on integrating data with different modalities, spatial resolutions, and error structures. For instance, ecosystem models may need to combine localized measurements, such as soil phosphorus at centimeter scales, with coarse climatic data at global resolutions. Without proper harmonization, such mismatches can inflate validation performance while misrepresenting ecological processes (Bonan and Doney, 2018; Kautz et al., 2018). Recent advances such as transfer learning and related meta-learning approaches seek to mitigate these limitations by drawing on information from data-rich regions or tasks to support predictions in data-sparse contexts (e.g., MetaFlux; Nathaniel et al., 2023), but their effectiveness still depends on the extent to which ecological patterns can be transferred across regions, and remains constrained by limited observations in underrepresented areas.

4.2. Scalability and generalization issues

The spatial transferability of ML models remains a major bottleneck in large-scale ecological applications. Models trained in local environments often show degraded performance when extrapolated to regional or global scales because ecological relationships are strongly shaped by site-specific factors such as microclimate and land-use history. For example, carbon flux models trained on local eddy covariance observations can capture local relationships among radiation, temperature, and carbon dioxide fluxes, yet their predictive accuracy often declines across biomes (Farahani and Goodwell, 2024). Similar limitations compromise ML derived large spatial-scale products such as SoilGrids, whose predictive performance deteriorates at the regional or local scales, sometimes yielding low correlations between predicted and observed soil organic carbon in heterogeneous regions (Guevara et al., 2018; Tifafi et al., 2018). Approaches including spatial cross-validation, domain adaptation, and macroecological constraints have been proposed to improve generalization, but global applications still require assumptions about unobserved drivers and the transferability of learned relationships (Ploton et al., 2020; Meyer and Pebesma, 2022; Stock et al., 2023). As a result, predictions across novel environmental gradients remain uncertain, and large-scale ecological maps are often better viewed as hypothesis-driven approximations than uniformly reliable representations of global patterns (Shi et al., 2022).

In the temporal dimension, scalability is further constrained by the breakdown of stationarity assumptions in ecological systems. Unlike many ML applications that operate under relatively stable statistical relationships, ecological forecasting is increasingly facing situations where the governing processes are themselves evolving. Climate change, land-use change, biological invasions, and increasing disturbance regimes continuously reshape ecosystem structure and function, creating environmental conditions that may have no historical analogue. Consequently, relationships learned from historical observations may no longer remain valid when projected into future scenarios, making extrapolation substantially more difficult than interpolation within observed data (Dietze et al., 2018; Reichstein et al., 2019). This challenge extends beyond distribution shift because environmental change can alter the ecological mechanisms linking predictors and responses, thereby violating one of the fundamental assumptions underlying supervised learning. For example, empirical relationships between vegetation productivity and climate derived from historical observations may become unreliable under future climate extremes or new environmental conditions (Dietze et al., 2018). Current efforts therefore increasingly combine adaptive learning strategies with process-informed modelling to improve robustness under changing environmental conditions, although reliable prediction under unprecedented future climates remains one of the major challenges for ecological ML.

4.3. Interpretability and causal inference

The increasing complexity of ML architectures is typically associated with reduced transparency. Methods such as deep CNN rely on multiple hidden layers and large numbers of interacting parameters to capture nonlinear environmental relationships, often functioning as black boxes. As a result, the ecological mechanisms underlying their outputs are difficult to interpret, which limits their value for scientific inference and can reduce stakeholder confidence in applications such as conservation management and policy development (Taskin et al., 2024).

To address issues related to predictive accuracy and ecological understanding, researchers increasingly use explainable artificial intelligence (XAI) methods. Post hoc attribution approaches trace feature contributions through deep neural networks, helping identify hidden spatial and temporal associations, generate ecological hypotheses, and provide partial insight into learned representations (Dramschr et al., 2025). However, these methods explain statistical model behaviour rather than ecological mechanisms. Recent efforts have therefore moved beyond post

hoc explanation toward scientifically interpretable machine learning, where model reasoning is explicitly aligned with domain knowledge and physical or ecological processes. For example, the Scientifically-Interpretable Reasoning Network (SciReN) demonstrates how hidden relationships in the carbon cycle can be discovered through interpretable reasoning rather than solely through feature attribution (Fan et al., 2026). Different model architectures trained on the same data may produce different explanations, particularly when predictors are strongly collinear in space or time (Rudin, 2019).

Ecological research ultimately seeks to understand the mechanisms governing natural systems rather than simply identifying correlative relationships. Building upon interpretable models, causal inference provides a framework for distinguishing genuine ecological mechanisms from spurious statistical associations. Integrating causal inference with ML has become an important direction for ecological research, particularly where management actions and environmental policies require estimates of intervention effects rather than interpolation of historical patterns. For example, combining causal ML with longitudinal observations has enabled researchers to estimate the contribution of biodiversity to ecosystem productivity across global grasslands while accounting for climatic and edaphic confounding factors (Dee et al., 2023). Nevertheless, robust causal inference remains challenging because ecological systems are characterized by non-stationarity, unmeasured confounders, and complex feedback loops that violate standard causal assumptions. Future progress will therefore require closer integration of causal inference frameworks, ecological process knowledge, and interpretable ML to ensure that inferred relationships reflect ecological mechanisms rather than statistical artifacts (Runge et al., 2019; Runge et al., 2023).

4.4. Uncertainties

Challenges in applying ML to ecology also arise from uncertainties that accumulate throughout the pipeline, from data acquisition to feature construction and model inference. Spatiotemporal heterogeneity creates pervasive sample bias and uneven detection probabilities in observational data, making it difficult to distinguish pseudo-absences from true absences in species distributions (Phillips et al., 2009). Flux tower and sensor observations are further affected by environmental noise, instrument drift, and microclimatic disturbances, which together create an irreducible baseline of stochastic uncertainty (Mauder et al., 2013). In addition, ecological variables are often misaligned across spatial and temporal scales, requiring interpolation, resampling, and gap filling. Although these procedures improve data usability, they also introduce smoothing and extrapolation bias. When variables are further derived from climate model outputs or remote sensing products, their inherent errors propagate into model inputs and accumulate through the analysis. Moreover, many of these “observational” products are themselves generated through process-based models, statistical retrieval algorithms, or data assimilation systems. Consequently, ML models trained on such products may partly learn assumptions and process understanding already embedded within the upstream models, rather than independent ecological relationships. This complicates the interpretation of model outputs and can make it difficult to distinguish genuinely new ecological insights from the propagation of existing model structures. Beyond uncertainties introduced during data acquisition and preprocessing, additional uncertainty arises from model selection and stochastic optimization during training, which can lead to differences in predictions across alternative model realizations.

Uncertainty quantification has therefore become a central issue in ecological ML, although its implementation remains limited by both methodological and practical constraints (Abdar et al., 2021). Existing approaches can be broadly divided into two groups. One follows a frequentist perspective, using methods such as Monte Carlo dropout or bootstrap resampling to approximate predictive distributions by perturbing models or data (Gal and Ghahramani, 2016;

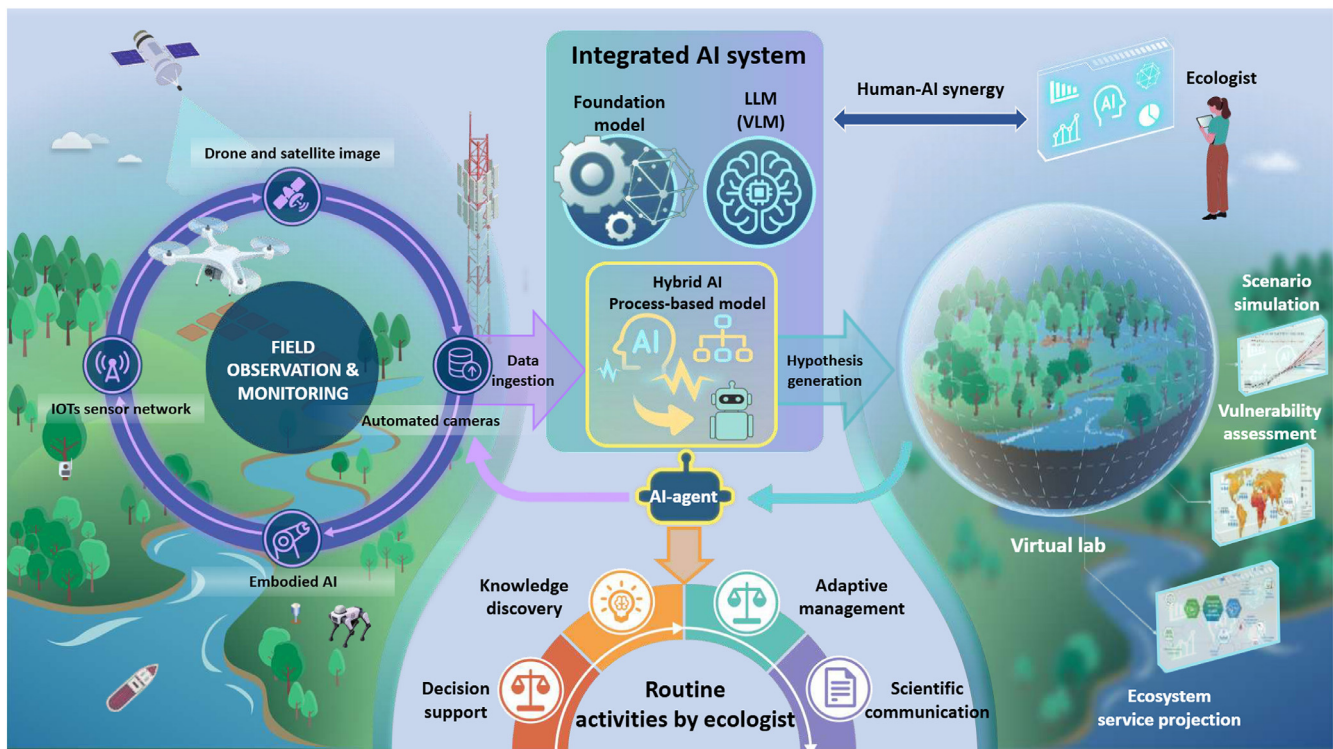


Fig. 3. Integrated Artificial Intelligence (AI) system architecture for future ecological study. The diagram depicts the closed-loop workflow of modern ecological research, integrating automated data acquisition, hybrid process-based AI modeling, and AI-assisted decision-making.

Lakshminarayanan et al., 2016). The other is based on Bayesian inference, representing epistemic uncertainty through posterior distributions of model parameters, for example via variational inference (Blundell et al., 2015). Frequentist approaches are often easier to implement, whereas Bayesian methods may provide more robust mean predictions, but both are constrained by computational cost and model complexity. More importantly, deep neural networks lack intrinsic mechanisms for representing uncertainty and often produce overconfident predictions, especially in high-dimensional, small-sample, or out-of-distribution settings. Although ensemble methods can improve predictive stability and help characterize uncertainty, multiscale ecological processes and internal variability still make uncertainty decomposition and validation difficult (Dietze et al., 2018).

5. Future prospects

Recent advances in AI and ML are creating new opportunities for analysing ecological systems and integrating increasingly large and heterogeneous environmental datasets. These developments come at a time when ecological research faces growing demands for spatially explicit, temporally resolved and decision-relevant predictions under rapid environmental change. However, progress in this area is unlikely to be driven by modelling skills alone, but also by the extent to which data acquisition, representation learning, and ecological process understanding can be tightly integrated. Despite widespread adoption, ML in ecology remains constrained by limitations in data quality, generalization, interpretability and, critically, causal inference. Recent work points towards a transition to more integrated analytical frameworks, although this transition remains uneven and largely confined to well-observed systems. With time, we envision a transformative shift in future ecological research, in which the role of ecologists will evolve from managing numerous fragmented tasks (such as field surveys, manual soil sampling, data organization, analysis, visualization, manuscript preparation, and reporting), toward a human–AI synergetic

paradigm (Fig. 3). In this AI-assisted working mode, coordinated systems of intelligent AI agents undertake routine and labor-intensive tasks, while ecologists take supervisory and interpretative roles, critically evaluating outputs, identifying potential errors, and providing iterative feedback to refine model performance and address system limitations (Fig. 3).

Ecological data collection is expanding from sparse field measurements to multiscale systems that combine in-situ sensors, unmanned aerial vehicles and satellite observations. These advances improve spatial coverage but do not fully resolve persistent biases arising from uneven sampling effort, accessibility constraints and taxonomic underrepresentation. As a result, many ecological datasets remain geographically and biologically imbalanced. With continued advances in next-generation sensing technologies and embodied AI, ecological observation systems may increasingly integrate AI-assisted sensing, automated quality control, anomaly detection, adaptive sampling, and field campaign planning. These developments will extend coverage across undersampled regions and environmental gradients, including deep-sea ecosystems, high-latitude permafrost zones, and dense forest canopies. AI-enabled and embodied platforms further enable adaptive sampling, whereby autonomous systems dynamically adjust sampling strategies in response to evolving observations, newly generated hypotheses, or novel understanding from hybrid-AI process-based modeling, substantially alleviating sample imbalance in both time, space and domain knowledge.

Evidence from Earth system science suggests that foundation models can capture large-scale spatiotemporal structure and support cross-task generalization, as demonstrated by geospatial models such as Prithvi, SatMAE and AlphaEarth (Jakubik et al., 2023; Kirillov et al., 2023; Brown et al., 2025), as well as data-driven forecasting systems including FourCastNet and Pangu-Weather (Pathak et al., 2022; Bi et al., 2023; Guo et al., 2024). However, ecological data differ from domains in which foundation models have been most successful, being sparse, taxonomically biased and unevenly distributed across space and time. These

characteristics limit scalability and complicate representation learning across ecosystems. Moreover, current models primarily encode statistical associations rather than causal processes, reducing their reliability for extrapolation under novel environmental conditions or for evaluating management interventions. Although ecological applications of foundation models remain at an early stage, recent developments suggest considerable potential for ecological research. The applications most likely to benefit largely from foundation models are those involving multimodal data integration, cross-region knowledge transfer, and large-scale ecological synthesis, where conventional models often struggle with heterogeneous data sources and limited observations. Future progress will depend not only on scaling model architectures and training datasets, but also on integrating ecological process knowledge and improving evaluation under out-of-distribution environmental conditions.

Meanwhile, LLMs and AI agents are substantially expanding capacities for data integration and analysis, enabling more comprehensive and efficient synthesis of increasingly complex ecological information. Such approaches can support tasks including literature synthesis, data extraction, parameter exploration, scenario analysis, hypothesis generation and testing, report and writing within structured workflows (Gougherty and Clipp, 2024; Gallois et al., 2025; Minasny et al., 2025; Lu et al., 2026). This has the potential to improve efficiency and reproducibility, particularly in evidence synthesis across large and fragmented document collections. However, current systems remain limited in quantitative reasoning, spatial analysis and uncertainty representation, and are prone to hallucination and the propagation of biases present in training data (Dorm et al., 2026). Their effective use therefore depends on domain-specific adaptation, explicit tool integration and continued human oversight. The substantial potential of such tools for ecological research will be further realized through ongoing advances in agentic architectures, tool-augmented reasoning, and the incorporation of process-informed constraints tailored to ecological systems.

Advances in observation, representation learning, workflow automation and reasoning point towards the emergence of integrated “virtual laboratories” for ecological studies. Here, we use this term to describe analytical environments that conduct a range of analytics, from data analyses (e.g., pattern recognition), process-based modelling, data assimilation, to scenario simulation. Such systems can enable more systematic exploration of scenario space. For example, they may support the evaluation of land-use impacts on carbon cycling or the analysis of ecosystem responses to extreme climatic events after policy shifts. The reliability of these systems remains strongly reliant on data quality, model structure and the representation of key ecological processes, and their outputs need to be rigorously evaluated against independent observations and experiments. Future AI-assisted workflows may progressively integrate observations, modelling, simulation, and decision support within unified analytical environments. However, these systems should be viewed as tools that augment ecological research rather than autonomous replacements for scientific investigation. More broadly, progress will depend on close collaboration between ecologists and AI researchers, together with continued efforts to improve data quality, causal understanding, interpretability, uncertainty quantification, and model validation. Rather than replacing ecological expertise, AI is expected to function primarily as a decision-support technology that augments human judgement. Ecological fieldwork, experimental design, and scientific responsibility will remain under human oversight, while AI increasingly supports data acquisition, analysis, and knowledge synthesis.

6. Conclusions

Machine learning is becoming an important component of ecological research, enabling the analysis of increasingly large and heterogeneous datasets and supporting applications ranging from identifica-

tion to spatiotemporal prediction, process understanding and conservation decision-making. These advances have been driven by rapid growth in ecological observations and developments in ML methods. However, their effectiveness remains constrained by several challenges, including data sparsity and bias, scale dependence, limited interpretability and weak links to underlying ecological processes, which restrict generalization and reduce confidence in predictions under novel conditions. Recent advances in next-generation observation systems, foundation models, large language models and AI agents are creating new opportunities for more integrated analytical frameworks in ecology, including adaptive observation, multimodal data integration and increasingly automated workflows. Together, these developments point towards an AI-assisted paradigm in which ecologists increasingly supervise AI-enabled systems for data acquisition, analysis and modelling, while retaining central roles in interpretation, validation and ecological reasoning. Ultimately, the impact of AI in ecology will depend on progress in data quality, causal understanding and model validation, with AI likely to extend rather than replace ecological expertise in supporting robust inference and decision-making.

CRediT authorship contribution statement

Jiapei Wu: Writing – original draft, Methodology, Data curation, Conceptualization. **Yiqi Luo:** Writing – review & editing, Formal analysis. **Fayong Liu:** Validation, Data curation. **Cuijuan Liao:** Formal analysis, Data curation. **Yike Wang:** Formal analysis, Data curation. **Honglin He:** Investigation, Data curation. **Dawei Tong:** Data curation. **Zhenqiang Liu:** Data curation. **Yuyang Zhao:** Data curation. **Xinjie Meng:** Data curation. **Caihui Sun:** Data curation. **Junzhi Zhou:** Methodology. **Jiang Zhang:** Methodology, Conceptualization. **Yuanyuan Huang:** Writing – review & editing, Formal analysis, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors did not use ChatGPT, AI-assisted technologies, or any other generative AI tools in the writing, editing, or preparation of the manuscript. The authors are solely responsible for the content of this publication.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data and code are available at Zenodo via a private link for peer review:

https://zenodo.org/records/19583494?preview=1&token=eyJhbGciOiJIUzUxMiJ9.eyJpZCI6IjI5MGZyZmMhLTZkMGU0NGI2NS1hOTIyLTcxZmYyNjExNmMyOSIsImRhdGEiOiOnt9LCJyYW5kb20iOiI2OTIyZGM4ZTU4YTlhOTBhZmMhNWUwZjczY2Y3MyJ9.L93T_nVMHAOb6z5luJ_h03vcKvHpb01khGYSjwk41CR88pnU16Xy7hZe_CsxQKWLanpZlqGzZCxCxVuhrXqEk_bQ. Details of the literature collection and data processing procedures used for Figs. 1 and 2 are provided in Supplementary materials.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.geosus.2026.100533](https://doi.org/10.1016/j.geosus.2026.100533).

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